ALPHA-BETA PRUNING ALGORITHM

AIM

To implement the **Alpha-Beta Pruning Algorithm** for game playing to find the optimal move while reducing the number of nodes evaluated in the Minimax search tree.

Algorithm

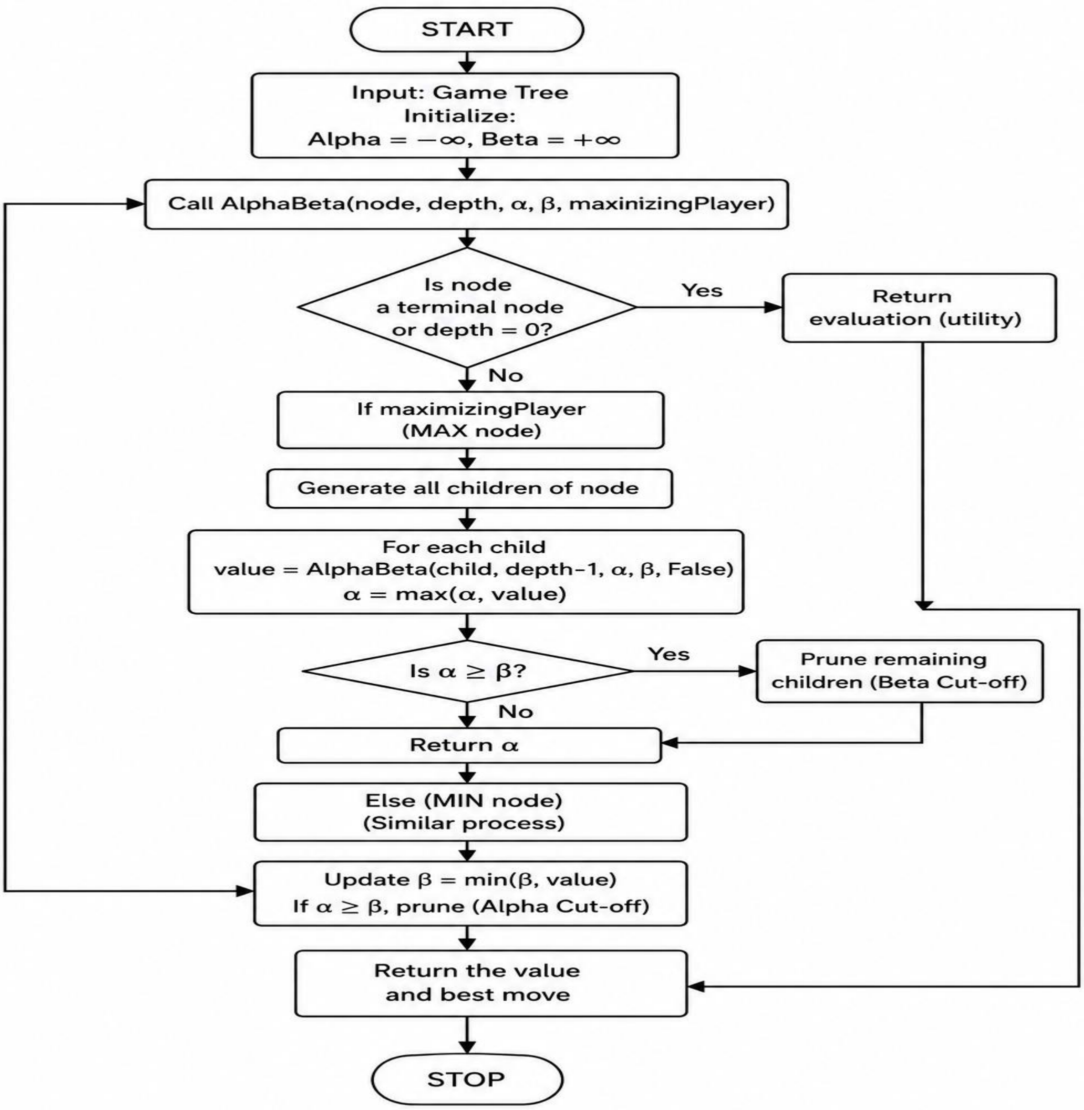
1. Start.
2. Initialize **Alpha = -∞** and **Beta = +∞**.
3. Traverse the game tree using the Minimax algorithm.
4. Update **Alpha** at MAX nodes and **Beta** at MIN nodes.
5. If **Alpha ≥ Beta**, prune the remaining branches.
6. Continue until the terminal nodes are reached.
7. Return the optimal move and its utility value.
8. Stop.

SAMPLE CODE

```
# Alpha-Beta Pruning Algorithm

MAX = 1000

MIN = -1000

# Alpha-Beta function

def alphabeta(depth, nodeIndex, maximizingPlayer, values, alpha, beta):

 # Terminal node

    if depth == 3:

        return values[nodeIndex]

 if maximizingPlayer:

        best = MIN

 for i in range(2):

            val = alphabeta(depth + 1, nodeIndex * 2 + i,

                            False, values, alpha, beta)
```

```
            best = max(best, val)
            alpha = max(alpha, best)
 # Beta Cut-off
            if beta <= alpha:
                break
 return best
else:
        best = MAX

        for i in range(2):
            val = alphabeta(depth + 1, nodeIndex * 2 + i,
                            True, values, alpha, beta)
            best = min(best, val)
            beta = min(beta, best)

            # Alpha Cut-off
            if beta <= alpha:
                break
return best
# Driver Code
if __name__ == "__main__":
 # Leaf node values
    values = [3, 5, 6, 9, 1, 2, 0, -1]
result = alphabeta(0, 0, True, values, MIN, MAX)
print("The optimal value is:", result)
```

FLOW CHART



OUTPUT

```
"C:\Users\Madhu Bavana\PycharmProjects\PythonProject1\.venv\Scripts\python.exe" "C:\Users\Madhu Bavana\
The optimal value is: 5

Process finished with exit code 0
```